b. Opportunistic fungi: When the immune status of the host is reduced, fungi will induce or cause infection.

c. Toxigenic fungi: Toxins produced by fungi are responsible for the illness or death of patients after ingestion of the contaminated food.

d. Allergenic fungi: Allergens are secreted by the fungi which cause allergic reaction in the human beings.

9.1.1 Mycoses

Diseases caused by the medically important fungi are called Mycoses. Mycoses are classified according to the specific site of involvement.

a. Superficial Mycoses: The infection is limited to the outer most layers of the skin and its appendages. Example: Malassezia and Piedra infection

b. Cutaneous Mycoses: The infection extends deeper into the epidermis and it also invades hair and nails. Example: Dermatophytoses.

c. Sub cutaneous Mycoses: The infection extends to dermis, subcutaneous tissue and muscles by any traumatic injury. Example: Mycetoma

d. Systemic Mycoses: The infection originates from lungs and later spreads systemically to other organs. Systemic mycoses along with the opportunistic fungal infection are known as deep mycoses. Example: Cryptococcosis

e. Opportunistic Mycoses: The infection occurs when the immune status of the individuals is altered. It is common among immune compromised and immune suppressed patients. Example: Candidiasis

Aeromycology

The Aeromycology is the study of air borne fungi, its types and the seasonal variations of allergenic fungal spores in the environment.

There are certain fungal pathogens which cause infections associated with workers in mycological laboratories. To avoid this safety procedures and equipments safety levels or bio safety levels (BSL) are used. BSL - 1 is used for low - risk microorganisms and BSL - 4 is used for highly risk pathogens.

Infobits

Medical Mycology in India

In India, the fungal infections are known since the ancient civilization mentioned in **Aryan documents such as Atharva Veda** Mycetoma is described as Padavamikam meaning ant hillfoot this was observed by John Gill in Madurai district of Tamil Nadu in 1842 which was designated as '**Madura foot**'.

9.1.2 Characteristics of Fungi

Fungi are heterotrophic organisms that exist as saprophytes, commensal or parasites. They are found on decaying vegetative matter and also in soil. Morphological features, cell structure, reproduction, nutritional requirement and thermal dimorphism in the pathogenic fungi are described as follows:

i. Morphological Features

Fungi are eukaryotic with well defined cell wall and internal membrane bound organelles. The cell wall is composed of polysaccharides and chitin. A fungus

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